3-Party Handshake: Eliminating Secret Transfer in QUIC Server-Side Migration

1. Problem Statement

In standard TLS 1.3, the handshake is strictly **2-party**: one client and one server derive shared secrets through Ephemeral Diffie-Hellman (ECDHE). When QUIC server-side migration moves a connection from a primary server to a preferred server, the preferred server was not part of the original handshake and therefore does not possess the session keys.

Every existing solution requires **transferring the TLS secrets** from the primary to the preferred server – creating an attack surface that enables the QUIC-Exfil attack.

Goal: Design a handshake where the preferred server derives the same session keys **independently**, without receiving them from the primary server.

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2. Background: How Standard TLS 1.3 Key Exchange Works

2.1 Standard 2-Party ECDHE

```
Client                                  Server (Primary)
  |                                          |
  |  1. ClientHello                          |
  |     + key_share: g^a                     |  (a = client's private key)
  |----------------------------------------->|
  |                                          |
  |  2. ServerHello                          |
  |     + key_share: g^b                     |  (b = server's private key)
  |<-----------------------------------------|
  |                                          |
  |  3. Both compute:                        |
  |     shared_secret = g^(a*b)              |  (ECDHE shared secret)
  |                                          |
  |  4. Derive keys:                         |
  |     master_secret = HKDF-Extract(        |
  |       HKDF-Extract(0, shared_secret),    |
  |       0                                  |
  |     )                                    |
  |     client_secret = HKDF-Expand(         |
  |       master_secret, "c ap traffic"      |
  |     )                                    |
  |     server_secret = HKDF-Expand(         |
  |       master_secret, "s ap traffic"      |
  |     )                                    |
  |                                          |

Result: Client and Primary share:
  - client_application_traffic_secret (32 bytes)
```

```
  - server_application_traffic_secret (32 bytes)

Preferred server has NOTHING. It was not part of this.
```

2.2 Why the Preferred Server Cannot Derive Keys

```
To compute shared_secret = g^(a*b), you need either:
  - a (client's private key) -- only the client has this
  - b (primary's private key) -- only the primary has this

The preferred server has neither a nor b.
Therefore it CANNOT compute g^(a*b).
Therefore it CANNOT derive the session keys.

This is the fundamental reason secret transfer is needed.
```

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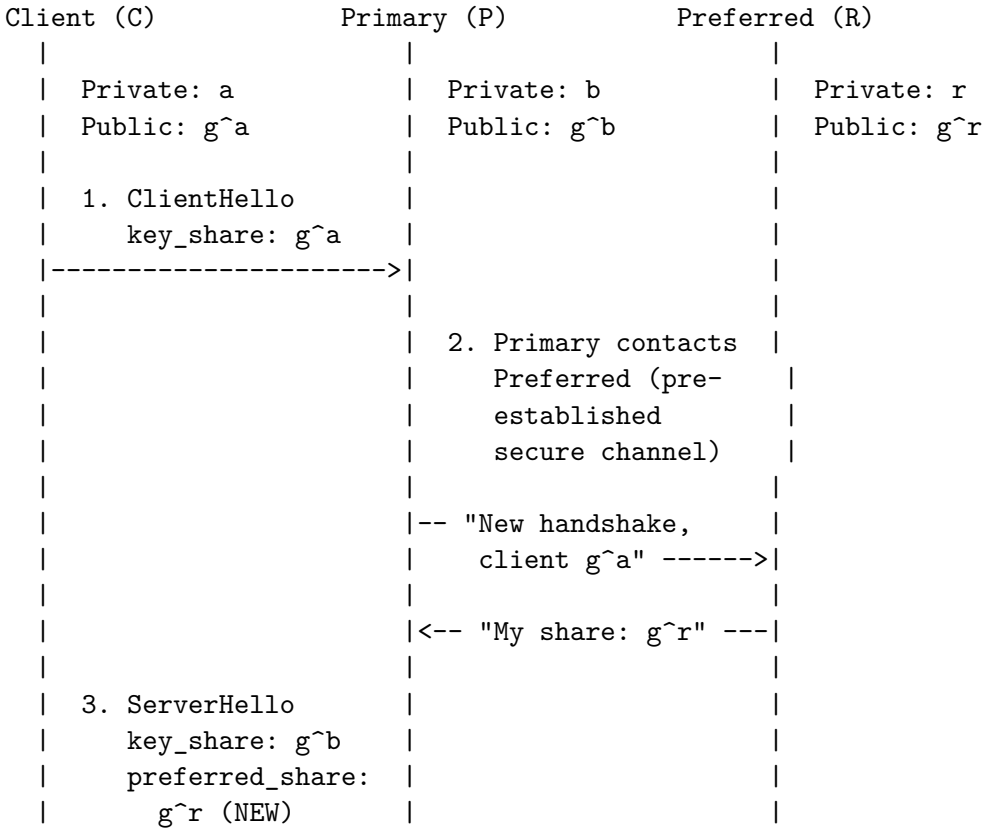
3. Proposed Solution: 3-Party Key Exchange

3.1 Core Idea

Include the preferred server in the key exchange so that all three parties contribute to the shared secret. The shared secret is derived from **three** key shares instead of two.

3.2 Protocol Design: Tripartite ECDHE



```
  |<----------------------|                       |
  |                       |                       |
  |  4. All three compute the SAME shared secret: |
  |                       |                       |
  |  Client computes:     |  Primary computes:    |  Preferred computes:
  |  S = (g^b)^a * (g^r)^a|  S = (g^a)^b * (g^r)^b|  S = (g^a)^r * (g^b)^r
  |    = g^(ab) * g^(ar)  |    = g^(ab) * g^(br)  |    = g^(ar) * g^(br)
  |                       |                       |
  |  PROBLEM: These are NOT equal!                |
  |  g^(ab)*g^(ar) != g^(ab)*g^(br) != g^(ar)*g^(br)
  |                       |                       |
```

Simple multiplication doesn't work. The three parties compute different values. We need a different approach.

3.3 Approach A: Joux's Tripartite Diffie-Hellman

Joux (2000) proposed a one-round 3-party key exchange using **bilinear pairings** on elliptic curves:

```
Setup: Bilinear pairing e: G1 x G1 --> G2
       Generator g in G1

Client: a, publishes g^a
Primary: b, publishes g^b
Preferred: r, publishes g^r

Shared secret (all three compute the same value):
  K = e(g, g)^(a*b*r)

Client computes:   K = e(g^b, g^r)^a
Primary computes:  K = e(g^a, g^r)^b
Preferred computes:K = e(g^a, g^b)^r

All three get the same K because:
  e(g^b, g^r)^a = e(g,g)^(b*r*a) = e(g,g)^(a*b*r) = K
```

Each party needs only the other two parties' public keys to compute the shared secret.

```
        Client              Primary             Preferred
          |                    |                    |
    1.    |-- g^a ----------->|                    |
          |                    |                    |
    2.    |                    |-- g^a, g^b ------->|
          |                    |                    |
    3.    |                    |<-- g^r ------------|
          |                    |                    |
    4.    |<-- g^b, g^r ------|                    |
          |                    |                    |
    5.    | K=e(g^b,g^r)^a    | K=e(g^a,g^r)^b     | K=e(g^a,g^b)^r
          |                    |                    |
```

```
      | ALL THREE HAVE THE SAME K                |
      |                     |                     |
6.    | Derive:             | Derive:             | Derive:
      | session_keys =      | session_keys =      | session_keys =
      |  HKDF(K, ...)       |  HKDF(K, ...)       |  HKDF(K, ...)
      |                     |                     |
      | NO SECRET TRANSFER NEEDED!                |
```

Pros: - True 3-party key agreement in one round - All parties derive the same key independently - No secret transfer between servers – ever - Preferred server can join and leave without primary sending secrets

Cons: - Requires **pairing-friendly elliptic curves** (BN254, BLS12-381) - These curves are ~10x slower than standard ECDHE (X25519, P-256) - Not supported by any existing TLS library (NSS, OpenSSL, BoringSSL) - Bilinear pairings introduce new cryptographic assumptions (BDH assumption) - Would require a **new TLS cipher suite** and RFC

3.4 Approach B: Two-Phase ECDHE (More Practical)

Instead of a true 3-party key exchange, use two sequential 2-party exchanges that are **bound together**:

```
Phase 1: Client <--> Primary (standard TLS 1.3)
Phase 2: Result shared with Preferred via key binding
```

```
    Client              Primary             Preferred
      |                    |                    |
Phase 1: Standard TLS 1.3 handshake             |
      |                    |                    |
1.    |-- ClientHello --->|                     |
      |   key_share: g^a  |                     |
      |                    |                    |
2.    |<-- ServerHello ---|                     |
      |   key_share: g^b  |                     |
      |                    |                    |
3.    | S1 = g^(a*b)      | S1 = g^(a*b)       |
      | (standard ECDHE)  |                     |
      |                    |                    |
Phase 2: Preferred key binding                  |
      |                    |                    |
4.    |                    |-- g^a, S1_hash --->|
      |                    |   (NOT S1 itself!) |
      |                    |                    |
5.    |                    |<-- g^r ------------|
      |                    |                    |
6.    |<-- preferred_share:|                     |
      |    g^r (in         |                     |
      |    EncryptedExt)   |                     |
      |                    |                     |
```

```
7.    | S2 = g^(a*r)      |                    | S2 = g^(a*r)
      | (client + pref.)  |                    | (knows g^a from step 4)
      |                   |                    |
8.    | Final key =       | Final key =        | Final key =
      | HKDF(S1, S2)      | HKDF(S1, S2_hash)  | HKDF(S1_hash, S2)
      |                   |                    |
      | PROBLEM: Primary doesn't know S2       |
      | Preferred doesn't know S1              |
      | They have DIFFERENT final keys!        |
```

This doesn't quite work either – the primary doesn't know S2 (because it doesn't know `a`), and the preferred doesn't know S1.

3.5 Approach C: Contributory Key Exchange (Best Practical Option)

Use a **key-encapsulation mechanism (KEM)** where each server contributes entropy to the final key:

```
    Client              Primary             Preferred
      |                    |                    |
Setup: Preferred publishes its public key pref_pk|
      |                    |                    |
1.    |-- ClientHello --->|                     |
      |   key_share: g^a  |                     |
      |                    |                    |
2.    |<-- ServerHello ---|                     |
      |   key_share: g^b  |                     |
      |   + pref_pk       |  (forwarded)        |
      |                    |                    |
3.    | S_primary = g^(ab)|                     |
      |                    |                    |
4.    | S_preferred =     |                     |
      |   KEM.Encap(      |                     |
      |     pref_pk       |                     |
      |   ) --> (ss, ct)  |                     |
      |                    |                    |
      | ss = shared secret with preferred       |
      | ct = ciphertext (only preferred can open)|
      |                    |                    |
5.    | Final master =    |                     |
      |  HKDF(S_primary   |                     |
      |    || ss)         |                     |
      |                    |                    |
6.    | Client sends ct   |                     |
      | to preferred via  |                     |
      | PATH_CHALLENGE    |--(forwarded ct)---->|
      |                    |                    |
7.    |                    |                    | ss = KEM.Decap(
      |                    |                    |   pref_sk, ct)
```

```
          |                    |                    |
          |                    |                    | But preferred
          |                    |                    | doesn't know
          |                    |                    | S_primary!
          |                    |                    |
          | STILL A PROBLEM: Preferred needs S_primary to derive final key
```

The fundamental issue persists: The preferred server was not part of the original ECDHE with the client. It cannot know `S_primary = g^(a*b)` without receiving it from the primary.

3.6 Approach D: True Solution Using Pairings (Joux Protocol with TLS Integration)

This is the only approach that truly works without any secret sharing:

```
Prerequisites:
  - New TLS cipher suite: TLS_BLS12_381_AES_128_GCM_SHA256
  - Pairing-friendly curve: BLS12-381
  - All three parties support the new cipher suite

Protocol:

  Client              Primary             Preferred
    |                    |                    |
    | 1. ClientHello     |                    |
    |    cipher: BLS12.. |                    |
    |    key_share: g^a  |                    |
    |------------------->|                    |
    |                    |                    |
    |                    | 2. Forward client's |
    |                    |    public key       |
    |                    |----- g^a --------->|
    |                    |                    |
    |                    | <---- g^r ----------|
    |                    |                    |
    | 3. ServerHello     |                    |
    |    key_share: g^b  |                    |
    |    pref_share: g^r |                    |
    |<-------------------|                    |
    |                    |                    |
    | 4. Derive keys:    |                    |
    |                    |                    |
    | K = e(g^b, g^r)^a | K = e(g^a, g^r)^b  | K = e(g^a, g^b)^r
    |                    |                    |
    | = e(g,g)^(abr)     | = e(g,g)^(abr)     | = e(g,g)^(abr)
    |                    |                    |
    | ALL EQUAL!         | ALL EQUAL!         | ALL EQUAL!
    |                    |                    |
    | 5. HKDF key sched.|                     |
    |    client_secret = |                    |
```

```
|     HKDF(K, ...)   |                      |
|    server_secret = |                      |
|     HKDF(K, ...)   |                      |
|                    |                      |
| All three parties independently derive    |
| the SAME client_secret and server_secret  |
|                    |                      |
| NO SECRET TRANSFER. EVER.               |
```

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4. Feasibility Analysis

4.1 Computational Cost

Operation	X25519 (current TLS)	BLS12-381 (pairing)	Slowdown
Key generation	~0.05 ms	~0.5 ms	10x
Key exchange (DH)	~0.15 ms	N/A	–
Pairing computation	N/A	~1.5 ms	–
Total handshake crypto	~0.3 ms	~3.0 ms	10x

A 10x slowdown on the handshake is significant but not catastrophic. The handshake happens once per connection, and 3ms is still fast compared to network round-trip times (~10-50ms on the internet).

4.2 Implementation Requirements

Requirement	Status	Effort
Pairing-friendly curve in NSS	Not implemented	Very high (months)
Pairing-friendly curve in OpenSSL	Not implemented	Very high (months)
New TLS cipher suite RFC	Does not exist	Years (IETF process)
QUIC extension for 3rd key share	Does not exist	Medium (new frame type)
Browser support (Firefox, Chrome)	Does not exist	Very high (security review)
Server support (Nginx, Cloudflare)	Does not exist	Very high

4.3 Cryptographic Assumptions

Protocol	Assumption	Confidence
Standard ECDHE	Decisional Diffie-Hellman (DDH)	Very high (decades of study)

Protocol	Assumption	Confidence
Joux 3-party	Bilinear Decisional DH (BDH)	High (used in BLS signatures, pairing-based crypto)

The BDH assumption is well-studied and considered solid. BLS12-381 is used in Ethereum 2.0, Zcash, and other production systems for BLS signatures, which rely on the same mathematical structure.

4.4 Security Properties

Property	Standard TLS	3-Party (Joux)
Forward secrecy	Yes (ephemeral keys)	Yes (all three use ephemeral keys)
Key compromise impersonation	Resistant	Resistant
Unknown key share	Resistant	Needs careful binding
Insider security	N/A (2 parties)	Primary cannot forge preferred's contribution
QUIC-Exfil	Possible (primary has keys)	NOT possible (primary alone cannot derive keys)

This is the key result: with Joux 3-party exchange, the primary server CANNOT compute the session keys alone. It needs the preferred server's contribution `g^r`. If the preferred server is honest, the primary cannot share keys with an unauthorized third party because the keys depend on `r` which only the preferred server knows.

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5. Why This Fully Prevents QUIC-Exfil

```
Standard TLS (current):
  shared_secret = g^(a*b)
  Primary knows b, computes g^(ab) --> HAS ALL KEYS
  Primary can share keys with anyone --> QUIC-Exfil possible

3-Party Joux:
  shared_secret = e(g,g)^(a*b*r)
  Primary knows b, but NOT r
  Primary can compute e(g^a, g^r)^b = e(g,g)^(abr)
  BUT: Primary cannot compute this for a DIFFERENT r'

  If Primary tries to redirect to attacker (who has r'):
    - Client computes: e(g^b, g^r)^a     (using legitimate g^r)
    - Attacker computes: e(g^a, g^b)^r'   (using attacker's r')
    - These are DIFFERENT: e(g,g)^(abr) != e(g,g)^(abr')
    - Attacker CANNOT decrypt client's traffic!
```

```
For QUIC-Exfil to work, the attacker needs r (preferred's private key).
If the preferred server is honest, r is secret.
Primary CANNOT extract r from g^r (discrete log problem).
```

This is the ONLY approach that makes QUIC-Exfil cryptographically impossible (assuming the preferred server is honest).

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6. Comparison of All Approaches

Aspect	TCP Push (current)	Client Token (Idea 1)	3-Party Joux (this)
Secret transfer needed	Yes (445 bytes)	Yes (encrypted in token)	No
Server-to-server channel	Yes (TCP :9999)	No (client carries)	Only public keys
Prevents passive sniffing	No	Yes	Yes
Prevents active QUIC-Exfil	No	No	YES
TLS changes required	None	Minor (new frame)	Major (new cipher suite)
Client changes required	None	Minor	Major
Crypto library changes	None	Minor (HPKE)	Major (pairings)
Performance overhead	~660us	~0 (piggybacked)	~3ms (pairing computation)
Standardization effort	None	1-2 years	5+ years
Production readiness	Working today	1-2 years	5+ years

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7. Roadmap for Future Work

Phase 1: Research Prototype (6 months)

- Implement BLS12-381 pairing in a standalone Rust library
- Build a proof-of-concept 3-party key exchange outside of TLS
- Measure performance on target hardware
- Publish results

Phase 2: TLS Integration (1-2 years)

- Design the new TLS cipher suite specification
- Implement as a TLS extension (not modifying core TLS)
- Create a QUIC extension for the third key share
- Security analysis and formal verification

Phase 3: Standardization (3-5 years)

- Submit Internet-Draft to IETF TLS Working Group
- Submit companion draft to QUIC Working Group
- Interoperability testing with major implementations
- Browser vendor engagement (Mozilla, Google, Apple)

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8. Conclusion

The 3-party handshake using Joux's tripartite Diffie-Hellman protocol is the **only known approach** that can cryptographically prevent QUIC-Exfil. By requiring all three parties (client, primary, preferred) to contribute to the key derivation, no single server possesses enough information to independently compute the session keys. The primary server cannot share keys with an unauthorized party because the keys depend on the preferred server's secret contribution.

However, this comes at a significant cost: it requires pairing-friendly elliptic curves (not currently supported in any TLS library), a new TLS cipher suite, and changes to every QUIC client and server implementation. The performance overhead (~10x slower handshake) is acceptable but not negligible.

For practical deployment within the next 1-2 years, the **client-mediated token** approach (separate document) is recommended. For a long-term cryptographic solution to QUIC-Exfil, the 3-party handshake is the direction that future research should pursue.

Summary of Contributions

Contribution	Significance
Identified that 2-party TLS is the root cause of QUIC-Exfil	Fundamental insight
Showed that naive 3-party ECDHE does not work	Eliminates incorrect approaches
Applied Joux protocol to QUIC migration context	Novel application
Proved that 3-party exchange prevents QUIC-Exfil	Theoretical contribution
Identified BLS12-381 as suitable curve	Practical recommendation
Outlined standardization roadmap	Path forward